

## ASSIGNMENT 1

### INSTRUCTIONS

- 1) ***THIS IS A GROUP WORK***
- 2) ***WORK IN THE GROUPS ASSIGNED***
- 3) ***PREPARE A VISUALLY APPEALING POWER-POINT PRESENTATION TO BE USED TO PRESENT YOUR POINTS IN CLASS.***
- 4) ***INCLUDE A SHORT VIDEO THAT SPEAKS TO YOUR SUBJECT MATTER. YOU CAN RECORD ONE AS A GROUP, OR GET ONE FROM ANY SOURCE.***
- 5) ***AN ORAL PRESENTATION BY ALL GROUP MEMBERS, WHICH SHOULD LAST NO MORE THAN 15 MINUTES TO START IN THE NEXT CLASS.***
- 6) ***AVOID PLAGIARISM AND COPY-PASTED WORK***
- 7) ***EACH MEMBER OF THE GROUP TO UPLOAD ONLY ONE DOCUMENT SAVED USING THE GROUP NAME***
- 8) ***DEADLINE FOR UPLOADING THE SUBMISSION IS ON 12TH MARCH 2026***
- 9) ***STRICTLY ADHERE TO THE INSTRUCTIONS TO AVOID NEGATIVE MARKING***

### QUESTIONS

| GROUP | QUESTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| A     | <ol style="list-style-type: none"><li>1. You are given a dataset of 1M homes with 80 features. Compare the trade-offs between using Ordinary Least Squares (OLS) and Ridge Regression when features are highly correlated.</li><li>2. A store has a sparse matrix of "User-Item" interactions. Describe how Singular Value Decomposition (SVD) can predict missing ratings.</li><li>3. Explain how a Denoising Autoencoder learns robust features compared to a standard Autoencoder.</li><li>4. Describe how Deep Q-Networks (DQN) use "Experience Replay" to break the correlation between consecutive training samples.</li></ol> |
| B     | <ol style="list-style-type: none"><li>1. A bank wants to predict loan defaults. Why might a Random Forest be preferable to a Support Vector Machine (SVM) regarding feature interpretability for regulatory compliance?</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                    |

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|          | <ol style="list-style-type: none"> <li>Using the Apriori Algorithm, explain how a "Support" threshold that is too low can lead to the "Combinatorial Explosion" problem.</li> <li>For a multi-label classification task (where an image can be both "Sunny" and "Beach"), why is Binary Cross-Entropy used instead of Categorical Cross-Entropy?</li> <li>Contrast Content-Based Filtering with Collaborative Filtering. Which suffers more from the "Cold Start" problem?</li> </ol>                                                                                                                                                                 |
| <b>C</b> | <ol style="list-style-type: none"> <li>In a rare disease detection model (0.1% prevalence), your model achieves 99.9% accuracy but 0% recall. Analyze the "Accuracy Paradox" and propose a new loss function.</li> <li>Describe the mathematical process of using PCA to reduce a 1024 X 1024 image while retaining 95% of the variance.</li> <li>A model performs perfectly in the lab but fails in production because the data distribution changed. Analyze this Data Drift scenario.</li> <li>A startup has 1 million unlabeled images and a budget to label only 1,000. How can Active Learning identify which 1,000 images to label?</li> </ol> |
| <b>D</b> | <ol style="list-style-type: none"> <li>Explain the impact of the Naive Bayes independence assumption when dealing with phrases like "Free Money" versus "Free" and "Money" as separate tokens.</li> <li>How can Spectral Clustering be used to identify communities in a graph where traditional distance metrics fail?</li> <li>For time-series data (stock prices), explain why standard K-Fold Cross-Validation is inappropriate and propose Time-Series Split</li> <li>For a news aggregator, why is Stochastic Gradient Descent (SGD) preferred over Batch Gradient Descent?</li> </ol>                                                          |
| <b>E</b> | <ol style="list-style-type: none"> <li>For a high-frequency trading bot, compare the inference latency of a 1000-tree Gradient Boosted Decision Tree (GBDT) versus a Logistic Regression model.</li> <li>In a biological taxonomy study, contrast Ward's Linkage with Single Linkage regarding their sensitivity to outliers.</li> </ol>                                                                                                                                                                                                                                                                                                              |

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|          | <ol style="list-style-type: none"> <li>3. In an airport security scanner, would you optimize for the F1-Score, or would you prioritize Recall at all costs? Justify your choice.</li> <li>4. In SVMs, explain how the RBF Kernel allows for linear separation in an infinite-dimensional space without actually computing the coordinates.</li> </ol>                                                                                                                                                                                                                                                                                                        |
| <b>F</b> | <ol style="list-style-type: none"> <li>1. A dataset has missing values in "Parental Income." Contrast the impact of Mean Imputation versus K-Nearest Neighbors (KNN) Imputation on model variance.</li> <li>2. You have 100 features. Explain the difference between Feature Selection (e.g., LASSO) and Feature Extraction (e.g., PCA).</li> <li>3. A hiring AI is found to favor male candidates. Identify if this is a result of Representation Bias or Historical Bias in the training set.</li> <li>4. Contrast Boosting (sequential) and Bagging (parallel). Which is more likely to overfit if the base learners are too complex</li> </ol>           |
| <b>G</b> | <ol style="list-style-type: none"> <li>1. When predicting claim amounts, why might you use a Poisson Regression or Tweedie Distribution instead of standard Linear Regression?</li> <li>2. In a 50-layer Vanilla RNN, gradients disappear during backpropagation. Explain how LSTM or GRU architectures solve this using "Gating Mechanisms."</li> <li>3. A deep learning model denies a medical insurance claim. How would you use SHAP or LIME to explain the decision to the patient?</li> <li>4. Explain how Self-Training (Pseudo-labeling) can leverage a small labeled dataset and a massive unlabeled dataset.</li> </ol>                            |
| <b>H</b> | <ol style="list-style-type: none"> <li>1. Compare One-vs-Rest (OvR) and One-vs-One (OvO) strategies for an SVM classifying 50 different species of birds.</li> <li>2. A CNN for facial recognition is failing to recognize faces that are slightly rotated. Discuss the role of Data Augmentation vs. Spatial Transformer Networks.</li> <li>3. In a highly imbalanced dataset (fraud detection), why is the Precision-Recall Curve more informative than the ROC Curve?</li> <li>4. Suppose you are building a classifier that identifies cats and dogs. You have a dataset of 3,000 images containing cats, dogs, or other objects (neither cat</li> </ol> |

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|          | <p>nor dog). You randomly split the data into a 2,500 image training set and a 500 image test set. After training your classifier for a while, you observe it performs well on the training images, but poorly on the test images. What is the possible explanation?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>I</b> | <ol style="list-style-type: none"> <li>1. Analyze how Cost-Sensitive Learning can be applied to a Logistic Regression model to prioritize high-value customers over low-value ones.</li> <li>2. You need to build a "Hotdog vs. Not Hotdog" app with only 500 images. Explain why fine-tuning ResNet-50 is better than training a small CNN from scratch.</li> <li>3. You see training loss decreasing while validation loss increases. List four specific techniques (from architecture to data) to halt this.</li> <li>4. Consider a dataset for house price prediction. A colleague suggests using a very deep neural network for this regression task. Discuss how feature importance techniques can help in stakeholder communication</li> </ol> |
| <b>J</b> | <ol style="list-style-type: none"> <li>1. You are using Polynomial Regression to model temperature. At degree <math>d=15</math>, the training error is near zero, but test error is massive. Propose three ways to fix this</li> <li>2. Compare Dropout and Batch Normalization. Can they be used together, or do they conflict in certain architectures?</li> <li>3. Use the bias-variance tradeoff to explain why an Ensemble of many high-variance models (Trees) leads to a low-variance model (Forest).</li> <li>4. Discuss the ethical and professional implications of using AutoML (like Google Cloud AutoML) versus manual feature engineering for a critical infrastructure project.</li> </ol>                                             |
| <b>K</b> | <ol style="list-style-type: none"> <li>1. You run K-Means on customer data, but the clusters are heavily elongated. Explain why Gaussian Mixture Models (GMM) might provide better "soft" clustering.</li> <li>2. Define "Mode Collapse" in the context of a GAN generating handwritten digits and suggest a remedy (e.g., Wasserstein GAN).</li> <li>3. A model predicts a 70% chance of rain, but it only rains 20% of those times. Discuss how to use Platt Scaling to calibrate probabilities.</li> </ol>                                                                                                                                                                                                                                         |

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|          | <p>4. An industrial plant wants to predict machine failures using vibration sensors, temperature logs, audio recordings, and maintenance history. Suggest an appropriate ML model (or combination of models) and justify your choice.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>L</b> | <ol style="list-style-type: none"> <li>1. You have 20,000 gene expressions. Compare the visual insights gained from Principal Component Analysis (PCA) versus t-SNE.</li> <li>2. Contrast the "Single Shot" approach of YOLO with the "Region-based" approach of Faster R-CNN in terms of speed vs. accuracy.</li> <li>3. Explain how Federated Learning allows a hospital to train a shared ML model without sharing sensitive patient records.</li> <li>4. Consider a dataset for house price prediction. A colleague suggests using a very deep neural network for this regression task. Describe reasons why a deep network may not be ideal for this task.</li> </ol>                                                                                                             |
| <b>M</b> | <ol style="list-style-type: none"> <li>1. Contrast the use of Isolation Forests versus Autoencoders for identifying localized power surges.</li> <li>2. Why did the Transformer architecture (Self-Attention) largely replace LSTMs for machine translation?</li> <li>3. In a Multi-Armed Bandit problem for digital ads, explain the Epsilon-Greedy strategy.</li> <li>4. Suppose you are building a classifier that identifies cats and dogs. You have a dataset of 3,000 images containing cats, dogs, or other objects (neither cat nor dog). You randomly split the data into a 2,500 image training set and a 500 image test set. Why is it important to "reserve" some images for the test dataset? (Why shouldn't we use all 3,000 images to train the classifier?)</li> </ol> |
| <b>N</b> | <ol style="list-style-type: none"> <li>1. A library wants to categorize 100,000 unsorted PDFs. Explain the process of using Latent Dirichlet Allocation (LDA) to find latent themes.</li> <li>2. Describe the difference between Grid Search, Random Search, and Bayesian Optimization for tuning a 10-layer Deep Neural Network.</li> <li>3. Define the "Bellman Equation" and its role in calculating the value of a state in Reinforcement Learning.</li> </ol>                                                                                                                                                                                                                                                                                                                     |

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|  | <p><b>4.</b> Determine whether the tasks described below involve supervised learning or unsupervised learning. For supervised learning problems, identify them as regression, classification, or probabilistic classification.</p> <p>a) Predict the risk of an accident at an intersection, given features such as the time of day and weather.</p> <p>b) Identify cars, bicyclists, and pedestrians in video taken by an autonomous vehicle's cameras.</p> <p>c) Determine the probability that there is a stop sign in an image.</p> <p>d) Generate new road scenarios (generate streets, place stop signs and intersections) for testing autonomous vehicles in a simulation.</p> |
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